
SECTION 1

What the data actually is

The source is the AgentFormer repository, folder `datasets/eth_ucy`.

AgentFormer keeps the benchmark in a leave-one-scene-out layout. For each scene we take the file that carries no `_train` and no `_val` suffix. That file is

the held-out test split, and the released checkpoint for that scene never saw it.

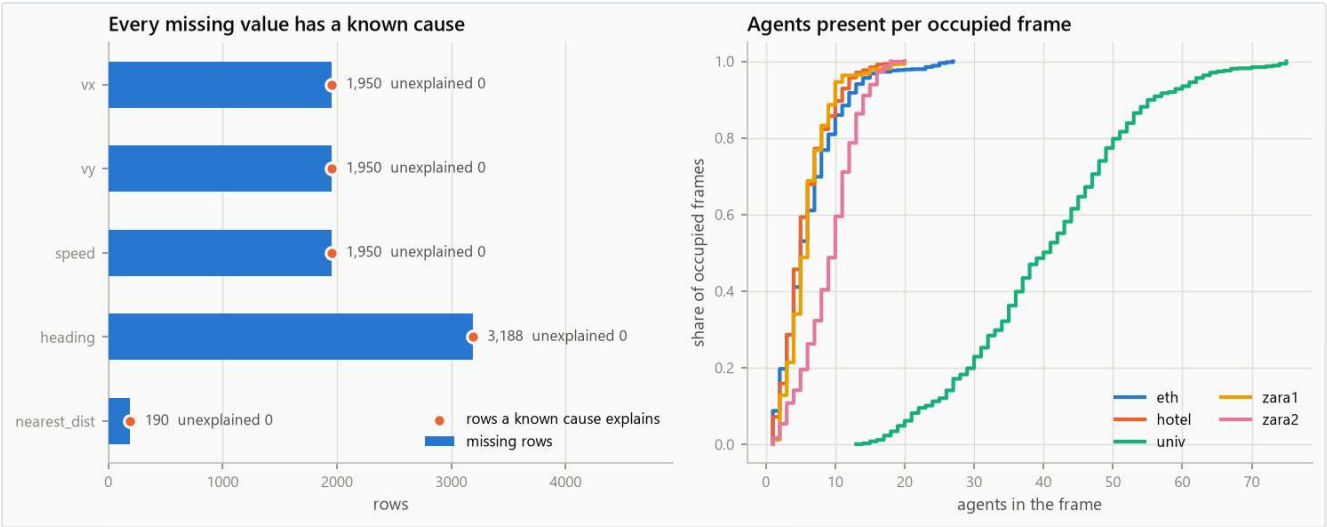
The text format holds 17 columns. Column 0 is the frame, column 1 the agent id, column 13 is x in metres and column 15 is y in metres. AgentFormer reads the same columns, so our positions and the model positions are the same numbers. The frame index already steps by 1 at 2.5 Hz, so no frame conversion is needed.

SCENE	SOURCE FILE OR FILES	ROWS	PEDESTRIANS
eth	eth/biwi_eth	5,492	360
hotel	hotel/biwi_hotel	6,543	389
univ	univ/students001 and univ/students003	39,766	849
zara1	zara1/crowds_zara01	5,153	148
zara2	zara2/crowds_zara02	9,722	204
Total	6 files	66,676	1,950

Scene sources. The scene univ is two separate recordings, which is finding F11.

The data is clean

Zero tracks skip a frame. Zero frame and agent pairs repeat. Every agent is a pedestrian. Every missing value has one of three known causes, and nothing is unexplained.



The first frame of a track has no earlier frame, so its velocity is missing. That accounts for 1,950 rows. A frame with a speed of exactly 0 has no heading, which adds 1,238 rows. An agent alone

SECTION 2

Three of the four proposal numbers hold

CLAIM IN THE PROPOSAL	RESULT	VERDICT
About 66,676 rows	66,676	Confirmed, exact
About 1,950 pedestrians	1,950	Confirmed, exact
univ holds about 60 percent of the rows	59.6 percent	Confirmed
About 2,416 windows	3,348 built, 2,841 eligible	Correct the text

The window count depends on the stride, and the proposal never states a stride. At a stride of 1 the pipeline builds 3,348 windows and 2,841 of them carry an ego. The number 2,416 matches no stride we tested. Replace the claim with the stride and the two counts.

A window covers 20 consecutive frames. An agent joins a window only when the agent is present in all 20 frames. A window is eligible when it holds 2 agents or more, because an ego needs at least 1 neighbour.

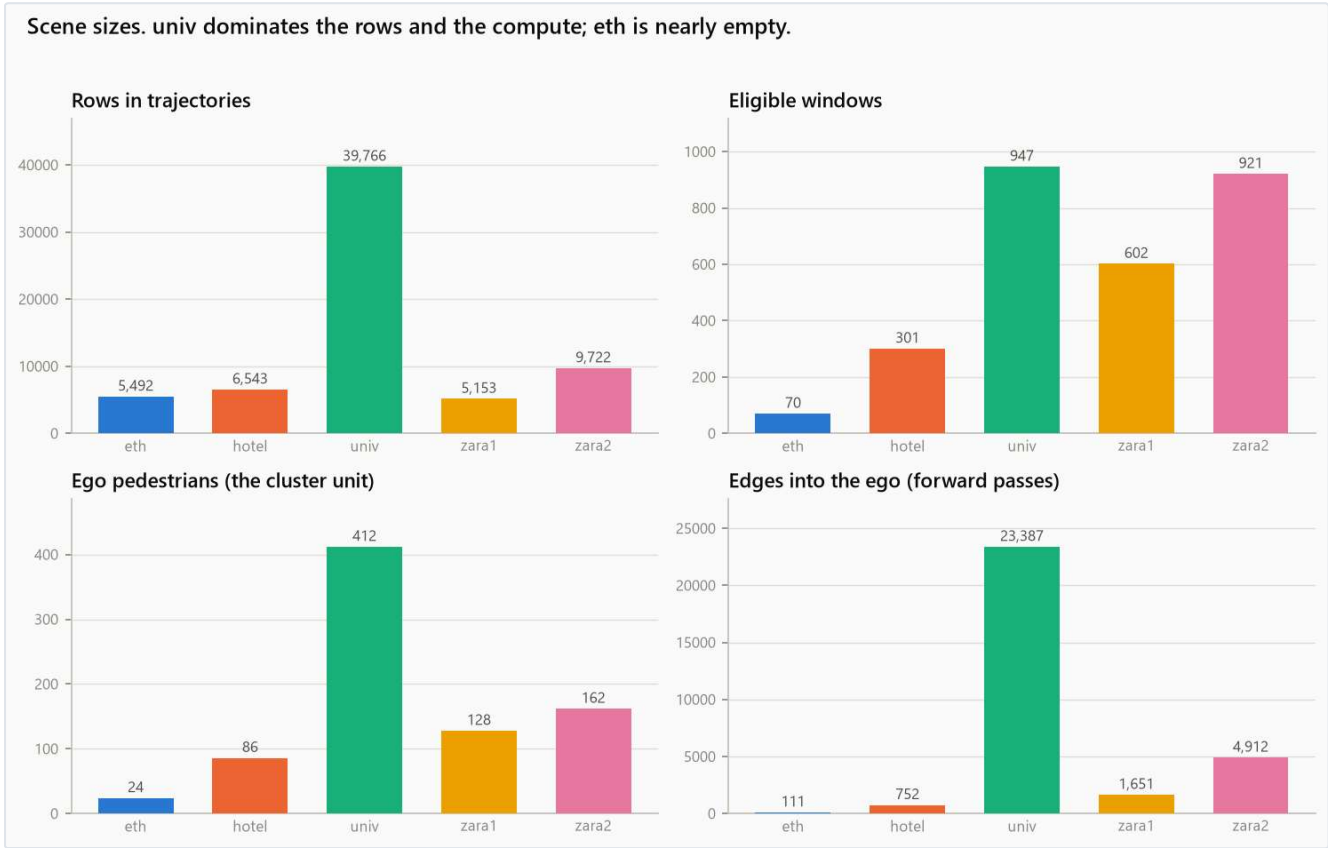
SECTION 3

The sample is two datasets wearing one name

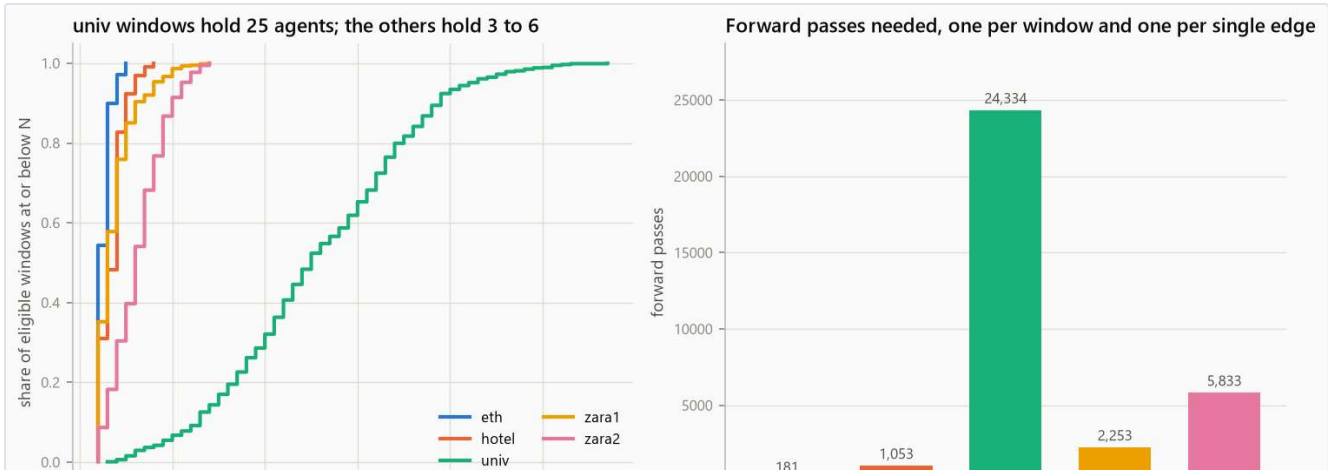
SCENE	ELIGIBLE WINDOWS	EGO PEDESTRIANS	MEAN N	MEDIAN N	MAX N	EDGES INTO THE EGO
eth	70	24	2.6	2	5	111
hotel	301	86	3.5	4	8	752

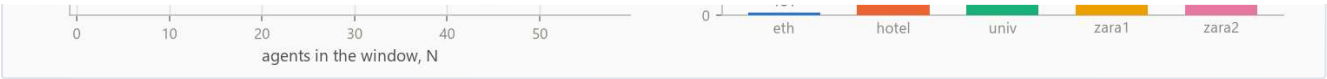
univ	947	412	25.7	25	57	23,387
zara1	602	128	3.7	3	14	1,651
zara2	921	162	6.3	6	14	4,912
Total	2,841	812	11.8	6	57	30,813

N is the agent count of a window. E is N minus 1, the number of edges that point into the ego.
E sets the cost of the brute force sweep, so the last column is the compute budget of the whole study.



The scene eth gives 70 eligible windows and 24 ego pedestrians. The scene univ gives 76 percent of the forward passes. Any pooled result is close to a univ result.





A univ window holds 25 agents. An eth, hotel or zara1 window holds 3 to 4. The two groups need different treatment, and finding F4 explains why.

SECTION 4

Eight findings that change the design

Settle every one of these before anybody runs an ablation. After `config.py` is hashed and committed, none of them can be changed without discarding results.

F1 The effective sample is between 195 and 846, not 2,841

Two windows that start 1 frame apart share 19 of their 20 frames. Many windows also share the same ego pedestrian. Both facts inflate the row count and neither adds information.

```
deff          = 1 + (mean_cluster_size - 1) * icc
n_effective = n_rows / deff
```

CLUSTER UNIT	CLUSTERS	MEAN SIZE	ICC	DEFF	EFFECTIVE N
Ego pedestrian	812	3.50	0.94	3.36	846
20-frame time block	190	14.95	0.97	14.57	195

The contract clusters on the pedestrian only. That choice misses the larger correlation. The time overlap costs more than the repeated pedestrian. The intraclass correlation above uses a proxy, because the shift does not exist until the ablation runs. The four proxies give a range from 0.56 to 0.94, and the table reports the largest, which is the safe one.

ACTION

Report both cluster units and take the smaller effective sample as the headline. Do not write

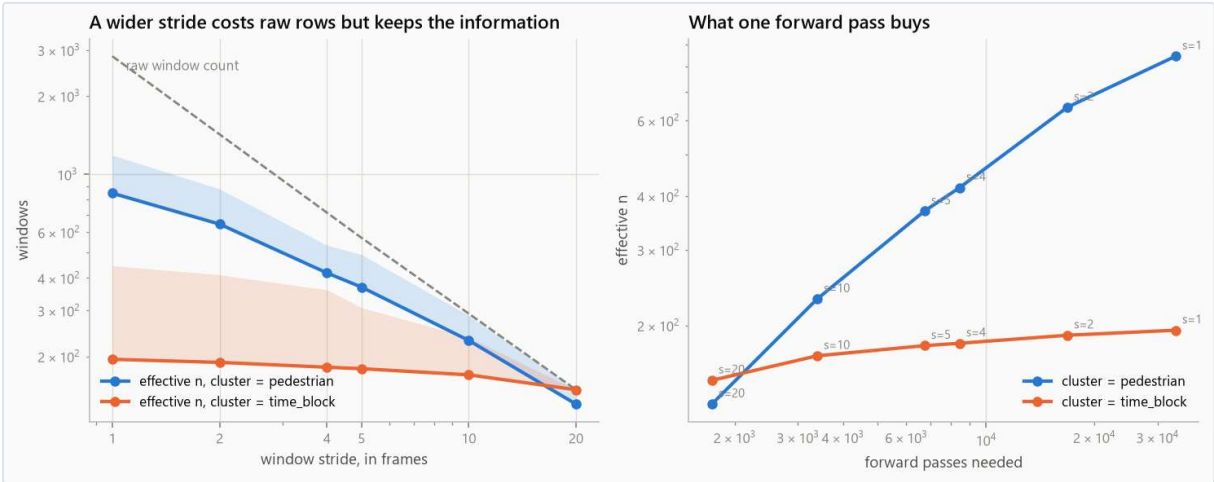
"n = 2,841" anywhere. Repeat this calculation on the real shift after the pilot ablation.

F2

A stride of 1 costs 20 times the compute and buys 31 percent more information

STRIDE	WINDOWS	FORWARD PASSES	EFFECTIVE N, PEDESTRIAN	EFFECTIVE N, TIME BLOCK	MEAN OVERLAP
1	2,841	33,654	846	195	1.00
2	1,424	16,836	644	190	1.00
4	713	8,440	418	182	1.00
5	571	6,741	369	179	1.00
10	292	3,390	230	170	1.00
20	149	1,728	131	149	0.00

One forward pass carries 20 latent draws. A stride of 1 therefore needs 673,080 sampled paths and a stride of 20 needs 34,560. At a stride of 20 the windows share no frame at all, so the time correlation disappears by construction.



The raw window count falls 19 fold from stride 1 to stride 20. The effective sample under time clustering falls from 195 to 149. Almost all of the extra rows at stride 1 are copies of rows we already hold.

ACTION

Freeze the stride in `src/hypotheses/design.py` before the first ablation run. A stride of 20

gives an honest design at 5 percent of the cost. A stride of 5 keeps 4 times more raw rows and still costs 5 times less than a stride of 1.

F3

The faithfulness index is undefined for 15 percent of the windows

The contract sets $FI = (\text{shift}(\text{morf}) - \text{floor}) / (\text{ceiling} - \text{floor})$, with **floor** the mean shift over the single edges and **ceiling** the largest. When E equals 1 the mean equals the maximum, so the denominator is 0 and FI is NaN. When E equals 2 the index can take only two values, so FI stops being continuous.

SCENE	WINDOWS	E = 1	SHARE	E = 1 OR 2	SHARE
eth	70	38	54%	63	90%
hotel	301	93	31%	145	48%
univ	947	0	0%	1	0%
zara1	602	212	35%	348	58%
zara2	921	80	9%	168	18%
Total	2,841	423	15%	725	26%

H3 regresses FI on the context. That regression therefore loses 15 percent of the rows, and it loses 90 percent of the eth rows. The remaining eth sample is 7 windows.

ACTION

Add a minimum E to the eligibility rule, or state the FI sample as a separate and smaller sample with its own count. Grace and Prem decide together, before the run.

F4

One removed edge is 63 percent of the eth graph and 4 percent of the univ graph

`PRIMARY_N_REMOVED` is 1. The dose of that treatment is not the same across scenes.

SCENE	MEAN E	ONE EDGE AS A SHARE OF THE GRAPH
eth	1.6	63 percent
hotel	2.5	40 percent
zara1	2.7	36 percent
zara2	5.3	19 percent
univ	24.7	4 percent

The same instruction removes most of the graph in eth and almost none of it in univ. A pooled test therefore mixes two different interventions. The scene fixed effect of H3 absorbs part of that difference, but H1 and H2 have no such term.

ACTION

Report H1 and H2 per scene as well as pooled. The `removed_mass` column already exists in `perturbation_curves`. Use it as the dose and say so in the report.

F5 The scene eth cannot support leave-one-scene-out

The scene eth gives 70 eligible windows and 24 ego pedestrians. At a stride of 20 it gives about 3 windows. The tracks of eth are short: only 44 of its 360 tracks reach 20 frames.

`src/hypotheses/validate.py` runs leave-one-scene-out and reads the spread as heterogeneity. An estimate from 24 clusters carries a very wide interval, so the spread will look large whatever the truth is.

ACTION

Report the cluster count beside every leave-one-scene-out estimate. State in advance that the eth estimate is descriptive and not evidence.

F6 The scene univ decides the pooled answer

SHARE HELD BY UNIV	VALUE
Rows	59.6 percent
Eligible windows	33.3 percent
Ego pedestrians	50.7 percent
Single-edge forward passes	75.9 percent

A pooled result is close to a univ result. The scene univ is also the scene where one edge is the smallest share of the graph, so it is the scene least likely to show an effect.

ACTION

Give the per-scene table the same weight as the pooled test in the report.

F7

drop_short_tracks is a no-op for the windows and a trap for the features

A track shorter than 20 frames cannot cover 20 consecutive frames, so such a track never joins a window. We ran the pipeline both ways. The eligible window count is 2,841 either way, and the mean N per scene is identical to 2 decimal places.

The filter is not harmless, because the order of operations matters. The columns `density` and `nearest_dist` count the pedestrians that were really there. If the filter runs before `add_kinematics`, those two columns lose real people.

SCENE	ROWS IN SHORT TRACKS	SHARE OF THE SCENE
eth	4,292	78 percent
hotel	3,028	46 percent
univ	1,714	4 percent
zara1	99	2 percent
zara2	221	2 percent

Total

9,354

14 percent

ACTION

Always add the kinematics before any drop. `scripts/run_all.py` does this and `config.DROP_SHORT_TRACKS` is False. Keep it False. The flag stays only so that a reviewer can measure the other policy.

F8 **overlap_fraction cannot discriminate at a stride of 1**

The contract defines `overlap_fraction` as the share of the 20 frames also covered by another window. At a stride of 1 the mean is 0.9975 and the median is 1.0. The column is constant, so it carries no information. It becomes exactly 0 at a stride of 20, where the windows are disjoint.

ACTION

Keep the column, because the contract fixes it. Take the design decision from the design effect of F1, never from this column.

SECTION 5

Four findings to
note and move past

F9 **Every measured input is right skewed, so expect branch 3b**

QUANTITY	SKEW	EXCESS KURTOSIS	IMPLIED BRANCH
Step length, metres	1.09	1.76	3b
Speed, metres per second	1.09	1.76	3b
Turn per frame, radians	7.23	68.57	3b
Nearest distance. metres	3.97	22.61	3b

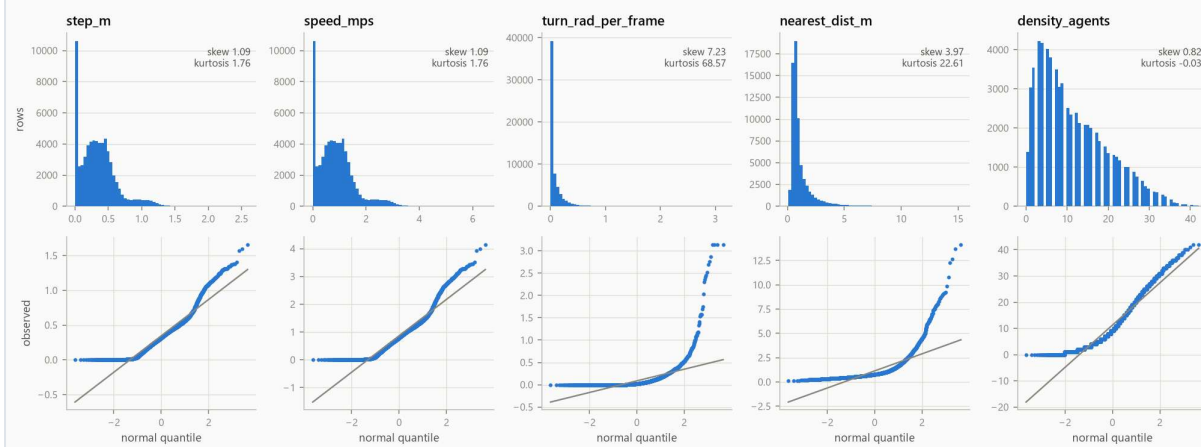
Local density, agents

0.82

-0.03

3b

Every quantity is right skewed. Expect branch 3b, not a t-test.



The shift of an ablation is a distance. A distance cannot be negative and it has a long right tail, so the same shape is very likely. The step length equals the speed times DT, so those two panels show an identity, not a second result.

This is evidence about the inputs, not about the paired difference D. The difference of two skewed variables can still be symmetric. `choose_test.py` must measure D itself. Do not choose the branch from this table.

F10 `weight_matched` is ill defined when E is small

`weight_matched_set` returns the smallest set of the lowest attention edges whose total attention reaches a target mass. When E is 1 there is no other edge, so the arm cannot run. When E is 2 the arm has one choice only. 26 percent of the windows have E of 2 or less.

ACTION

Define the fallback in `src/ablation/arms.py`. Log every window where the target mass is out of reach, and report the count.

F11 The scene univ is two separate recordings

The scene univ comes from `students001` and `students003`. Both files start at frame 0 and both use agent ids from 1 upward, so the two recordings are not one timeline. `src/data/loader_ethucv.py` adds an offset of 100 000 to the

One line in `src/data_loader_utils.py` adds an offset of 100,000 to the frame index and to the agent id of the second file. The offset is much larger than one window, so no window can span the two recordings. A test guards this.

ACTION

Add one line to `CONTRACT.md` section 3 that states the offset.

F12

At `n_removed` of 1, four arms are already inside the single-edge sweep

At a removal count of 1 the arms `morf`, `lerf`, `nearest` and `random` each remove exactly one edge. The brute force sweep of `src/faithfulness/index.py` already removes every edge on its own, so the four arms are lookups into a table that already exists.

The marginal cost of H1 and H2 at the primary setting is therefore zero once the sweep has run. This is the one finding that makes the budget easier.

ACTION

Build the sweep first. Read the four arms out of it.

SECTION 6

What honest power looks like

The table gives the minimum detectable effect of a one-sided sign test at an alpha of 0.05. The effect is the share of windows where MoRF gives the larger shift. A share of 0.5 is the null.

DESIGN	EFFECTIVE N	POWER 0.80	POWER 0.90
Stride 1, cluster by time block	195	0.589	0.604
Stride 1, cluster by pedestrian	846	0.543	0.550

Stride 5, cluster by time block	179	0.592	0.608
Stride 20, disjoint windows	149	0.601	0.618
Every window independent (wrong)	2,841	0.523	0.527

Read it this way. With the honest design we can detect “MoRF wins in 59 of every 100 windows”. We cannot detect “MoRF wins in 53 of every 100”. The naive count of 2,841 promises a precision the data does not hold.

The next decision belongs to Prem

`src/hypotheses/design.py` chooses the stride, chooses `MINIMUM_EFFECT` and `POWER` , writes all three into `config.py` , then prints `config.config_hash()` and commits the hash in the same commit. After that commit, nobody changes a number.

```
WINDOW_STRIDE      = 1      # freeze this in design.py first
MINIMUM_EFFECT     = None   # fill from the table above
POWER              = None   # fill from the table above
DROP_SHORT_TRACKS  = False  # keep it False. See F7
```

SECTION 7

What is built and who takes it next

Done and tested

STAGE	MODULE
1	scripts/download_ethucy.py
2	src/data/ — schema, loader, clean, windows, eligibility, synthetic
3	src/eda/ — summary, density, distributions, scaling, missing, overlap
3	scripts/run_all.py

Reproduce every number on this page with two commands.

```
python scripts/download_ethucy.py
python scripts/run_all.py
```

The four branches

PERSON	MODULES	GPU
Amogh	models, attention, ablation, scripts	Yes
Catherine	features, metrics	No
Grace	faithfulness, ekf	No
Prem	stats, hypotheses	No

Amogh ships `base.py` , `mock.py` and `cv.py` on day 1, because those three files unblock Grace. Prem ships `design.py` on day 1, because that file freezes `config.py` and everybody waits for the frozen numbers.

The full prompt for each branch is in `docs/BRANCH_PROMPTS.md` .

The largest schedule risk

The AgentFormer checkpoints and the AgentFormer code pin an old PyTorch. That risk blocks stages 7 to 9 and it belongs to one person. Prove that one checkpoint loads and returns a path in week 1, before anybody polishes anything else. Every other stage runs against `MockPredictor` from day 1, which is the whole point of the model seam.